

Combined Quantum and Post-Quantum Security for Earth-Satellite Channels - 2nd Part

Anju Rani et al. (IEEE International Conference on Quantum Communications, Networking, and Computing 2025)

July 16, 2025

Outline

1 Post-Processing

2 Emulation

3 Discussion

Outline

1 Post-Processing

2 Emulation

3 Discussion

Overview (First Part)

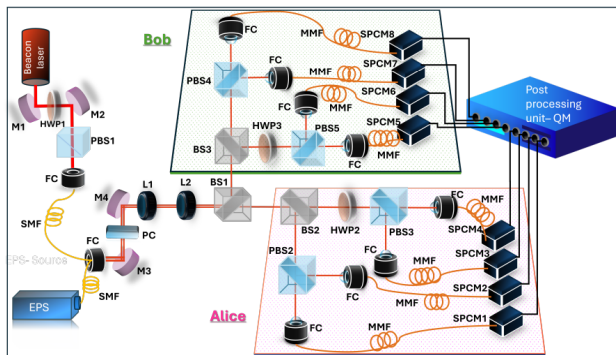


Fig. 2. The experimental setup for the BBM92 protocol. EPS: Entangled photon source; M: Mirror; HWP: Half-wave plate; PBS: Polarizing beam splitter; FC: Fiber coupler; SMF: Single-mode fiber; BS: Beam splitter; PC: Polarization correction optics which is a combination of a quarter-wave plate - half-wave plate - quarter-wave plate; L: Lens; MMF: Multi-mode fiber; SPCM: Single-photon counting module; QM: Quantum machine (post-processing unit).

Overview (Post-Processing)

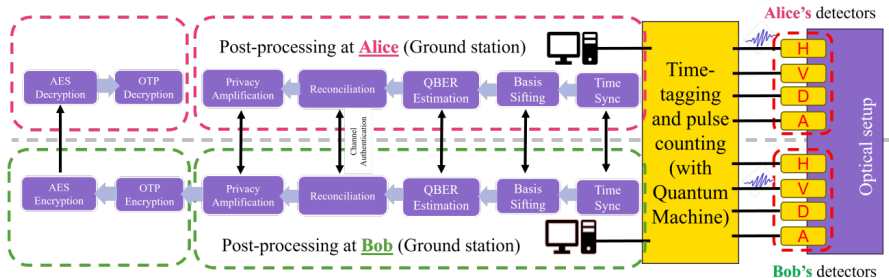


Fig. 3. A high-level diagram for the QKD-PQC post-processing hardware setup.

Step (a): Photon Time-Tagging

- During each detection window (50 ns), all 8 SPCMs are monitored:
 - Alice: 4 detectors (H, V, D, A) $\rightarrow T_a = [t_1^a, t_2^a, t_3^a, t_4^a]$
 - Bob: 4 detectors (H, V, D, A) $\rightarrow T_b = [t_1^b, t_2^b, t_3^b, t_4^b]$
- Example valid T_a (one detector clicked):

$$T_a = [-1, 12, -1, -1]$$

- Example invalid T_a (multiple clicks — noise):

$$T_a = [11, 14, -1, -1]$$

- If no photon detected:

$$T_a = [-1, -1, -1, -1]$$

Step (b): Time-Domain Filtering

- After collecting T_a and T_b , Alice & Bob exchange their arrays and filter events.
- Conditions for valid coincidence event:
 - 1 Exactly one positive time-tag in T_a and one in T_b .
 - 2 Time difference between them $\leq$ coincidence window t_c (1 ns in this experiment).
- Example:

$$T_a = [-1, 12, -1, -1], \quad T_b = [-1, -1, 11, -1]$$

- $|12 - 11| = 1 \leq t_c \rightarrow$ valid event
- If T_a or T_b has > 1 positive or time difference $> t_c \rightarrow$ discard

Step (c): Bit Mapping — Concept

- After time-domain filtering, each valid event identifies:
 - Which detector at Alice fired (H, V, D, A).
 - Which detector at Bob fired (H, V, D, A).
- Each outcome maps to two bits:
 - **Key bit (k)**: the raw bit value.
 - **Basis bit (b)**: which basis was measured.
 - H/V basis $\rightarrow b = 0$
 - D/A basis $\rightarrow b = 1$

Alice:

Outcome	k_a	b_a
H	0	0
V	1	0
D	1	1
A	0	1

Bob:

Outcome	k_b	b_b
H	1	0
V	0	0
D	0	1
A	1	1

Step (c): Bit Mapping — Example Event

Example:

- At Alice:

$$T_a = [-1, 12, -1, -1] \quad (\text{Detector 2 fired} \rightarrow \text{Outcome: } V) \\ \Rightarrow k_a = 1, \quad b_a = 0$$

- At Bob:

$$T_b = [-1, -1, 11, -1] \quad (\text{Detector 3 fired} \rightarrow \text{Outcome: } D) \\ \Rightarrow k_b = 0, \quad b_b = 1$$

Decision:

- If $b_a = b_b$, keep the event for the raw key.
- If $b_a \neq b_b$, discard the event.

In this example:

$$b_a = 0, \quad b_b = 1 \quad (\text{Different bases} \rightarrow \text{discard})$$

Repeat until $N_{\text{raw}} = 40,000$ valid bits are collected.

Raw Key Generation Process

- For each valid detection event (per detection window), extract:
 - At Alice: (k_a, b_a) — 2 bits per event.
 - At Bob: (k_b, b_b) — 2 bits per event.
- Decision:
 - If $b_a = b_b$, keep k_a and k_b as raw key bits.
 - If $b_a \neq b_b$, discard the event.
- Repeat over many detection windows to accumulate $N_{raw} = 40,000$ valid key bits.
- Final raw keys:

$$k_a = [k_a^{(1)}, k_a^{(2)}, \dots, k_a^{(40,000)}]$$

$$k_b = [k_b^{(1)}, k_b^{(2)}, \dots, k_b^{(40,000)}]$$

Step (d): Base Sifting

- After bit mapping:
 - Alice has: key bits k_a and basis bits b_a
 - Bob has: key bits k_b and basis bits b_b
- In this step:
 - Alice and Bob exchange their b_a and b_b (basis choices) over the classical channel.
 - They compare basis choices for each position:

$$b_a^{(i)} \stackrel{?}{=} b_b^{(i)}$$

- If $b_a^{(i)} \neq b_b^{(i)}$, they discard $k_a^{(i)}$ and $k_b^{(i)}$.
 - If $b_a^{(i)} = b_b^{(i)}$, they keep $k_a^{(i)}$ and $k_b^{(i)}$ for the raw key.
- Since basis choices are random (50% chance), about half of the bits are discarded.
- Remaining key length after this step:

$$N' = \frac{N_{\text{raw}}}{2}$$

Step (d): Base Sifting — Example

Scenario:

- After bit mapping, Alice and Bob have:

$$k_a = [1, 0, 1, 0, 1, 0], \quad b_a = [0, 1, 0, 1, 0, 1]$$

$$k_b = [0, 1, 1, 0, 0, 1], \quad b_b = [1, 1, 0, 0, 0, 1]$$

Step: Compare b_a and b_b element-wise.

Index	1	2	3	4	5	6
b_a	0	1	0	1	0	1
b_b	1	1	0	0	0	1
Keep?	✗	✓	✓	✗	✓	✓

Result:

- Kept bits:

$$k_a = [0, 1, 1, 0], \quad k_b = [1, 1, 0, 1]$$

at indices 2, 3, 5, 6.

- Remaining length:

$$N' \approx \frac{N_{\text{raw}}}{2}$$

Step (e): QBER Estimation — Concept

Goal: Estimate the Quantum Bit Error Rate (QBER) — the fraction of mismatched bits between Alice and Bob.

- After base sifting, Alice and Bob have N' bits.
- Alice randomly selects $N'/2$ bits and their indices.
- She sends the selected bits & indices to Bob over the classical channel.
- Bob compares these bits with his k_b at the same indices.
- QBER is calculated as:

$$Q = \frac{\# \text{ of mismatched bits}}{\text{total bits compared}}$$

Action:

- If Q exceeds a security threshold $\rightarrow$ abort session.
- Otherwise $\rightarrow$ continue to error correction.
- Bits used for QBER estimation are discarded.

Remaining key length after this step:

$$N = N' - \frac{N'}{2} = \frac{N'}{2}$$

Step (e): QBER Estimation — Example

Scenario:

- After base sifting:

$$N' = 8, \quad k_a = [1, 0, 1, 1, 0, 1, 0, 1], \quad k_b = [1, 1, 1, 0, 0, 1, 1, 1]$$

- Alice randomly selects $N'/2 = 4$ bits at indices [2,4,6,8]:

$$k_a^{\text{sample}} = [0, 1, 1, 1], \quad k_b^{\text{sample}} = [1, 0, 1, 1]$$

Comparison Table:

Index	k_a	k_b	Match?
2	0	1	No
4	1	0	No
6	1	1	Yes
8	1	1	Yes

QBER:

$$Q = \frac{2}{4} = 0.5$$

If $Q > \text{threshold} \rightarrow \text{abort}$, otherwise $\rightarrow \text{proceed}$.

Step (f): Error Correction — Concept

Goal: Correct any remaining discrepancies between k_a and k_b and synchronize their keys.

- After QBER estimation:

$$N = \frac{N'}{2}$$

- Alice and Bob use an LDPC code with a parity-check matrix $H_{M \times N}$ to correct errors.
- H is constructed as:
 - Step 1: Optimize degree distributions based on QKD parameters (R_c) and [23].
 - Step 2: Build H with Progressive Edge Growth (PEG) algorithm [24].

At Bob:

- Compute syndrome:

$$s_b = H \cdot k_b$$

- Encrypt s_b with AES: $\mathcal{E}_p(s_b)$.
- Compute Wegman–Carter MAC tag T over s_b .
- Send $\mathcal{E}_p(s_b) || T$ to Alice.

Step (f): Error Correction — Example & Outcome

At Alice:

- Receive $\mathcal{E}_p(s_b) || T$.
- Decrypt s_b using AES.
- Verify T with Wegman–Carter MAC.
- If MAC verification fails $\rightarrow$ abort.
- Otherwise, run belief-propagation decoder:

$$\hat{k}_a \text{ such that } H \cdot \hat{k}_a = s_b$$

Outcome:

- If decoding succeeds $\rightarrow$ Alice's $\hat{k}_a = k_b$.
- If decoding fails $\rightarrow$ abort and restart.

Step (g): Privacy Amplification — Concept

Goal: Remove any partial information an eavesdropper (Eve) may have gained, producing a shorter, secure final key.

- After error correction:
 - Alice has $\hat{k}_a$, Bob has k_b , each of length N .
 - Assume $\hat{k}_a = k_b$.
- Estimate secure key rate per bit:

$$r = 1 - h(Q) - \text{leak}_{EC}$$

where:

- Q : QBER estimated earlier.
- $h(x)$: binary entropy, $h(x) = -x \log_2 x - (1-x) \log_2 (1-x)$.
- leak_{EC} : info leaked during error correction.
- Final key length:

$$|k'| = \lceil r \cdot N \rceil$$

Step (g): Privacy Amplification — Implementation

Implementation: Use universal hashing with a Toeplitz matrix to compress k_a and k_b into secure keys.

- Alice generates a random Toeplitz matrix T of size:

$$\lceil rN \rceil \times N$$

- Both Alice and Bob compute:

$$k'_a = T \cdot \hat{k}_a, \quad k'_b = T \cdot k_b$$

- Final keys k'_a and k'_b are shorter but:
 - Fully secure against any info Eve might have.
 - Identical at Alice and Bob.
- Alice sends T to Bob over the classical channel.

Result: Short, secure, identical QKD keys at both sides:

$$k'_a = k'_b$$

Step (h): Encryption & Decryption — Concept

Goal: Use the final QKD key k' to encrypt and decrypt messages securely.

- After privacy amplification:

$$k'_a = k'_b$$

is the shared secret key of length $|k'|$.

- Alice wants to send data Z to Bob:
 - If $|Z| \leq |k'|$, use One-Time Pad (OTP) + AES.
 - If $|Z| > |k'|$, skip OTP and use AES only.
- OTP provides information-theoretic security if $|k'| \geq |Z|$.
- AES provides computational security.

Why both?

- If enough QKD key is available $\rightarrow$ use OTP.
- Otherwise $\rightarrow$ AES ensures secure encryption even for longer messages.

Step (h): Encryption & Decryption — Equations

Case 1: $|Z| \leq |k'|$ (use OTP + AES)

- Alice encrypts:

$$\begin{aligned}\mathcal{E}_q(Z) &= Z \oplus k'_a \\ Z' &= \mathcal{E}_p(\mathcal{E}_q(Z), k_p)\end{aligned}$$

where k_p is the pre-shared AES key.

Bob decrypts:

$$Z = \mathcal{D}_q(\mathcal{D}_p(Z', k_p), k'_b)$$

Case 2: $|Z| > |k'|$ (AES only)

- Alice encrypts:

$$Z' = \mathcal{E}_p(Z, k_p)$$

- Bob decrypts:

$$Z = \mathcal{D}_p(Z', k_p)$$

Notes:

- Encryption/decryption times are negligible compared to QKD key accumulation.

Outline

1 Post-Processing

2 Emulation

3 Discussion

Goal: Model the laboratory QKD experiment as a realistic LEO satellite-to-ground scenario.

- EPS is assumed to be on a **LEO satellite**.
- EPS distributes entangled photon pairs:
 - One photon $\rightarrow$ Alice (Ground Station 1)
 - One photon $\rightarrow$ Bob (Ground Station 2)
- Downlink is preferred over uplink:
 - Most propagation occurs in vacuum.
 - Less attenuation compared to uplink (ground $\rightarrow$ satellite).
- Loss mechanisms:
 - Absorption, scattering, diffraction.
 - Pointing loss & atmospheric turbulence.

Channel Loss Model:

$$A_{\text{link}} = A_{\text{atm}} + A_{fs}$$

- A_{atm} : Atmospheric absorption & scattering (1 dB)
- A_{fs} : Diffraction, telescope losses, pointing loss:

$$A_{fs} = 10 \log_{10} \left(\frac{L^2 \lambda^2}{D_T^2 D_R^2 T_T T_R (1 - P_l)} \right)$$

Parameters:

- Wavelength: $\lambda = 800 \text{ nm}$
- Telescope efficiencies: $T_T = T_R = 0.8$
- Pointing loss: $P_l = 0.2$
- Total link loss fixed: $A_{\text{link}} = 10.3 \text{ dB}$

Where does 10.3dB come from? — Reality Check

In the lab:

- Laboratory QKD setup: fiber + 1.5m free space.
- Measured loss:

$$A_{\text{link, lab}} = A_{\text{channel}} + A_{\text{detectors}} = 4.3 \text{ dB} + 6.0 \text{ dB} = \mathbf{10.3 \text{ dB}}$$

- Controlled conditions: perfect alignment, no weather, no turbulence.

Logical Build-up of the Earth–Satellite Emulation

Goal: Emulate a realistic LEO–to–ground QKD link in a laboratory setting.

- 1 **Target scenario:** LEO satellite equipped with EPS sends entangled photons to two ground stations.
- 2 **Known realistic link loss:** From prior literature & experiments, a typical LEO–ground link exhibits:

$$A_{\text{link}} \approx 10.3 \text{ dB}$$

(including both channel & detector losses).

- 3 **Break down A_{link} :** Use:

$$A_{\text{link}} = A_{\text{atm}} + A_{fs}$$

with $A_{\text{atm}} = 1.0 \text{ dB}$ (good weather).

- 4 **Model A_{fs} :** Loss due to diffraction, telescope inefficiencies, pointing error:

$$A_{fs} = 10 \log_{10} \left(\frac{L^2 \lambda^2}{D_T^2 D_R^2 T_T T_R (1 - P_I)} \right)$$

- 5 **Fix other parameters:** previous slide
- 6 **Explore feasible telescopes:** For fixed $A_{\text{link}} = 10.3 \text{ dB}$, they varied D_T , D_R , and computed L to see what telescope sizes correspond to realistic distances.

Result #1

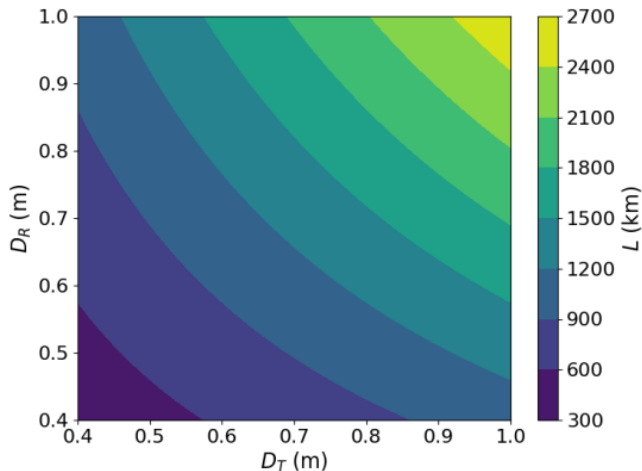


Fig. 4. Plot of link distance L as a function of transmitter (D_T) and receiver (D_R) telescope diameters for fixed link loss of 10.3 dB, $\lambda = 800$ nm, $P_l = 0.2$, and $T_T = T_R = 0.8$.

Prototype:

- Real-time QKD system integrated with PQC.
- Laboratory environment: channel length = 1.5 *m* (fiber + free-space).

Measured Channel Loss:

- Used 633 *nm* beacon laser.
- Measured transmittance: 0.61 $\rightarrow$ corresponds to 4.3 *dB* channel loss.

Objective: Test system performance at different EPS source powers:

- Measure **Visibility** (entanglement quality).
- Calculate **QBER** (bit error rate).
- Compute **Secure Key Rate**.

#1 Table I — Experimental Results

Measured values at different EPS source powers:

Source Power (mW)	Visibility	QBER	Secure Key Rate (kbps)
2.5	0.960 ± 0.009	0.016 ± 0.005	1.6
3.5	0.873 ± 0.019	0.063 ± 0.009	0.7
4.5	0.742 ± 0.013	0.136 ± 0.008	Nil

Observation:

- Best performance at 2.5 mW.
- Secure key rate drops to 0 when QBER exceeds 11%.

Results — Visibility, QBER, and Key Rate

Visibility:

- Highest at low EPS power (best entanglement quality).
- Decreases as EPS power increases due to noise & multi-photon events.

QBER:

- Related to Visibility: $QBER = \frac{1 - \text{Visibility}}{2}$.
- Low QBER ($< 11\%$) $\rightarrow$ secure key extraction possible.
- Higher EPS power $\rightarrow$ higher QBER $\rightarrow$ degraded security.

Secure Key Rate:

- Highest at 2.5 mW $\rightarrow 1.6\text{ kbps}$.
- Declines at higher EPS power.
- Becomes **Nil** when $QBER > 11\%$.

#2 Stability Test at 3.5 mW

Objective: Check the stability of QKD performance over time at a moderate EPS power.

Setup:

- EPS power fixed at 3.5 *mW*.
- Monitored QBER while accumulating 1000 sifted key bits.
- Also tested larger block sizes in full implementation.

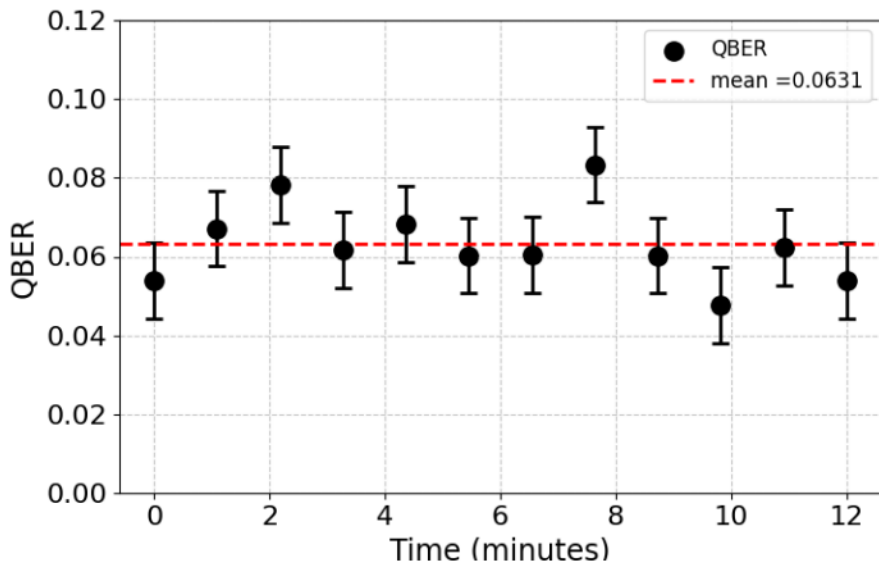
Observation:

- QBER remained stable and below the 11% threshold.
- Background counts and overall system were consistent across runs.

Conclusion:

- The experimental setup is stable and reliable at 3.5 *mW*.

Fig: Stable QBER



#3 Integration of QKD and PQC

Next Step:

- Used QKD-generated keys in a hybrid QKD + PQC encryption scheme.
- Applied to both:
 - Data payloads
 - Syndrome messages (for error correction)

Performance:

- Tested encryption & decryption on various data sizes.
- AES-256 processing time was negligible compared to QKD key generation.
- Confirmed real-time operation of the combined system.

Conclusion:

- Additional AES-256 layer adds security without significant overhead.
- The full QKD+PQC system is practical and stable.

Outline

1 Post-Processing

2 Emulation

3 Discussion

What Does “Additional Security” Mean?

Case 1: QKD is perfect (ideal implementation)

- QKD already provides information-theoretic security.
- AES-256 adds no additional benefit in this case.

Case 2: QKD is imperfect (realistic scenario)

- QKD can suffer from:
 - Side-channel attacks.
 - Poor hardware or implementation errors.
 - Satellite-specific vulnerabilities.
- Here, AES-256 provides fallback (computational) security.

Why AES-256?

- Secure even against quantum attacks (assuming Grover's limit $\rightarrow$ 128-bit).
- Acts as a safety net when QKD is compromised.

Why This Approach is Different

Other hybrid QKD+PQC systems:

- Combine QKD & AES keys into a single hybrid key.
- Require equal-length keys & composability proofs.
- Both primitives must be trusted at the same time.

This system:

- Layers QKD & AES-256 without combining keys.
- Applies AES-256 to data & reconciliation messages.
- Avoids composability issues & increases flexibility.

Trade-off:

- Encrypting syndrome messages with AES saves QKD bits & increases rate.
- But full information-theoretic security only holds if QKD is trusted.

Conclusion:

- AES-256 protects against practical attacks on QKD.
- The system remains robust even if one primitive is compromised.

Thank you for your time!